Unit V
Unit Name: Testing of Hypothesis

Overview:
Test of significance: Large sample test for single proportion, Large sample test for difference of proportions, For Single mean, For difference of means. Small Sample test, paired t test, Chi square test.

Outcome:
After completion of this unit, students would be able to:
1. solve problems involving testing of hypothesis
2. Analyze data samples using statistical methods.

Testing of Hypothesis

Z test: (Test of significance for large samples i.e. $n > 30$)

1.Testing of Significance for single mean:

a) $z = \dfrac{\overline{x} - \mu}{\sigma/\sqrt{n}}$ where σ is the standard deviation of the population.

b) If σ is not known, we use $z = \dfrac{\overline{x} - \mu}{s/\sqrt{n}}$ where s is the standard deviation of the sample.

Where $\overline{x}$ is sample mean, μ is the population mean, n is the sample size.

2.Test of significance for difference of means of two large samples:

a) $z = \dfrac{\overline{x_1} - \overline{x_2}}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$

Where $\overline{x_1}$ be the mean of a sample size n_1 from a population with mean μ_1 and variance σ_1^2 , $\overline{x_2}$ be the mean of a sample size n_2 from a population with mean μ_2 and variance σ_2^2 .

b) Under the null hypothesis that the samples are drawn from the same population where $\sigma_1 = \sigma_2 = \sigma$ i.e. $\mu_1 = \mu_2$ then $z = \dfrac{\overline{x_1} - \overline{x_2}}{\sigma\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$

c) If σ_1, σ_2 are not known and $\sigma_1 \neq \sigma_2$ then $z = \dfrac{\overline{x_1} - \overline{x_2}}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$

3.Test of significance for single proportion:

a) $z = \dfrac{p - P}{\sqrt{PQ/n}}$

Where $p = \dfrac{X}{n}$ is the observed proportion of success, X be the number of successes in n independent trials, P probability of success for each trial and $Q = 1 - P$ is the probability of failure.

4.Test of significance for difference of proportions:

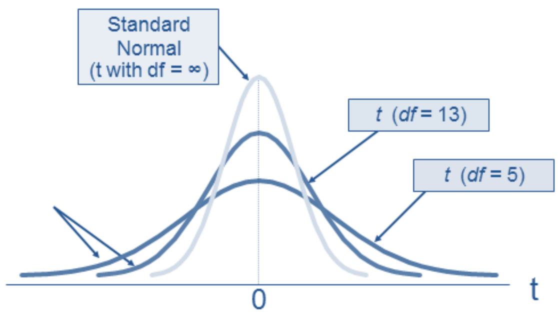
a) $z = \dfrac{\overline{p_1} - \overline{p_2}}{\sqrt{PQ\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$, $P = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \dfrac{X_1 + X_2}{n_1 + n_2}$ and $Q = 1 - P$

Where X_1 and X_2 are two samples of sizes n_1 and n_2, $p_1 = \dfrac{X_1}{n_1}$ and $p_2 = \dfrac{X_2}{n_2}$ are the sample proportions.

Small Samples

The sample is regarded as small if size of sample is less than 30.

Student's t-distribution



A random variable T is said to follow t-distribution if its probability density function is given by

$$f(t) = \frac{1}{\sqrt{v}\,\beta\left(\frac{v}{2}, \frac{1}{2}\right)}\left(1 + \frac{t^2}{v}\right)^{-(v+1)/2} ; -\infty < t < \infty$$

Where v denotes the number of degrees of freedom of the t-distribution.

Degree of freedom

The degrees of freedom in a statistical calculation represent how many values involved in a calculation have the freedom to vary.

Test 1: Test of significance of the difference between sample mean and population mean.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} \text{ with } \nu = n - 1.$$

Test 2: Test of significance of the difference between means of two small samples drawn from same population.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \text{ with } \nu = (n_1 + n_2 - 2).$$

Case i: If $n_1 = n_2 = n$ and if the samples are independent(not related)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}} \text{ with } \nu = 2n - 2.$$

Case ii: If $n_1 = n_2 = n$ and if the samples are associated in some way(or correlated)

Assume $H_0 : \bar{d}\left(= \bar{x} - \bar{y}\right) = 0$

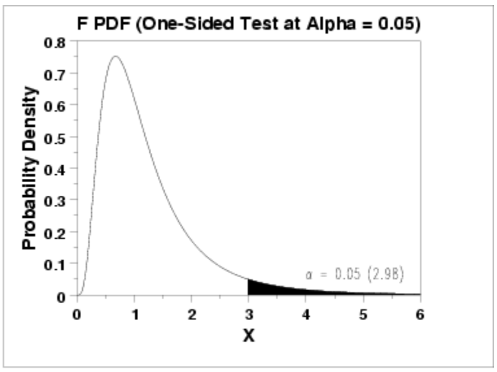
Solve using test statistic $t = \frac{\bar{d}}{s/\sqrt{n-1}}$ with $\nu = n - 1$.

Where $d_i = x_i - y_i, \bar{d} = \bar{x} - \bar{y};$ and std. dev. of d given by $s = \frac{1}{n}\sum_{i=1}^{n}\left(d_i - \bar{d}\right)^2$.

Test for ratio of variances (F-test using F-Distribution)

F-Distribution

The F distribution is a right-skewed distribution used most commonly in Analysis of Variance.



Test statistic $F = \dfrac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2}$ with $v_1 = n_1 - 1, v_2 = n_2 - 1$.

Here, $\hat{\sigma}_1^2 = \dfrac{n_1 s_1^2}{n_1 - 1}; \hat{\sigma}_2^2 = \dfrac{n_2 s_2^2}{n_2 - 1}$

Note: 1. F-test is not a two-tailed test and is always a right-tailed test, since F cannot be negative.

2. We should always make F > 1. This is done by taking the larger of the two estimates of $\hat{\sigma}^2$ as $\hat{\sigma}_1^2$ and by assuming that the corresponding degree of freedom as v_1

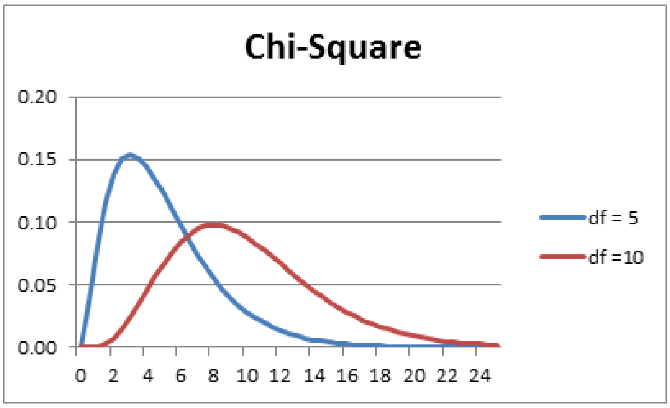
3. To test if two small samples have been drawn from the same normal population, it is not enough to test if their means differ significantly or not, because in this test we assumed that the two samples came from the same population or from populations with equal variance. So, before applying the 1-test for the significance of the difference of two sample means, we should satisfy ourselves about the equality of the population variances by F-test.

4. There are different tables for different level of significances.

Chi – Square Distribution

The sum of squares of n standard normal variables is defined to be a chi – square variable with *n* d.f.

$\chi^2 = X_1^2 + X_2^2 + \ldots X_n^2$



The χ^2 variable can be used to perform a variety of tests.

Uses of χ^2 test / distribution

1. **To test the independence of two or more attributes**:

χ^2 test can be used to determine whether there is any association between two or more attributes (characteristics) like color of eyes of mothers and daughters, heights of fathers and sons etc.

In these tests, we proceed on the null hypothesis that the attributes are independent i.e. there is *no association* between the attributes.

$$\chi^2 = \sum\left[\frac{(O_i - E_i)^2}{E_i}\right]$$

Where O_i = Observed frequency

E_i = Expected frequency

$\sum O_i = \sum E_i$

It follows χ^2 - distribution with n-1 degrees of freedoms.

2. **TEST OF INDEPENDENCE OF ATTRIBUTES-CONTIGENCY TABLES**

Consider two attributes A & B. Let A be divided into r classes $A_1, A_{2\ldots}$, A_r and B be divided into s classes $B_1, B_{2\ldots}$, B_s.

	A	**A_1**	**A_2**	**A_3**	**…..**	**A_r**	**Total**
B							
B_1		**(A_1B_1)**	**(A_2B_1)**	**(A_3B_1)**		**(A_rB_1)**	**(B_1)**
B_2		**(A_1B_2)**	**(A_2B_2)**	**(A_3B_2)**		**(A_rB_2)**	**(B_2)**
B_3							
…							
…		**…**	**…**	**…**	**…**	**…**	**…**
B_s		**(A_1B_s)**	**(A_2B_s)**	**(A_3B_s)**		**(A_rB_s)**	**(B_s)**
Total		**(A_1)**	**(A_2)**	**(A_3)**		**(A_r)**	**N**

Such a classification in which attributes are divided into more than two classes is known as *manifold classification*.

The various cell frequencies can be expressed in a table known as *r x s* manifold contingency table where (Ai) is the number of persons possessing attribute Ai, I = 1,2,…,r and (Bj) is the number of persons possessing attribute Bj, j = 1,2,…,s

Also $\sum_{i=1}^{r} A_i = \sum_{j=1}^{s} B_j = N$ Where N is total frequency.

Now, the expected number of persons possessing both the attributes Ai Bj is given by:

(under H_0) $Expected\ frequency = \dfrac{Row\ total \times column\ total}{overall\ total\ (N)}$

[The RHS can be obtained from the given table]

For large N,

$$\chi^2_{cal} = \sum_i \sum_j \left(O_{ij} - E_{ij}\right)^2 \sim \chi^2_{(r-1)(s-1)} d.f$$

Where O_{ij} and E_{ij} are the observed and expected frequencies respectively in the $(i,j)^{th}$ cell. (r and s are the number of rows and columns in the table respectively.

If $\chi^2_{cal} > \chi^2_{(r-1)(s-1)}$ at α LOS, then we reject H_0 and conclude that the attributes are not

independent. Otherwise we have no reason to reject H_0. We obtain $\chi^2_{(r-1)(s-1)}$ at α LOS from the chi-squared table, and it is obtained in the same manner as reading the t distribution. All the areas given in the table are the right tail. Since the question is about the attributes being dependent or independent, we don't have to worry about one-tailed and two-tailed tests.
While testing the independence of attributes, the Null Hypothesis H_0 will always be that the attributes are independent. The Alternative Hypothesis will suggest that they are dependent, i.e. there is some relationship between them.

References of the entire unit

1: Probability, Statistics and Random Processes, T. Veerarajan, Tata McGraw Hill, 3rd edition.

2: Applied Mathematics, G.V. Khumbhojkar, C. Jamnadas & Co.

Large Sample Test:

Tests for Mean:

1. Test of Significance of the difference between sample mean and population mean.

Problems:

1) A sample of 100 students is taken from a large population. The mean height of the students in this sample is 160 cm. Can it be reasonably regarded that, in the population, the mean height is 165cm, and the SD is 10cm?

   [Ans: $z_{cal} = -5$. H_0 is rejected at 1% level of significance.(It is not statistically correct to assume that $\mu = 165$.)]

2) The mean breaking strength of the cables supplied by a manufacturer is 1800, with an SD of 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cable has increased. To test this claim, a sample of 50 cables is tested and it is found that the mean breaking strength is 1850. Can we support the claim at 1% LOS?

   [Ans: $z_{cal} = 3.54$. H_0 is rejected at 1% level of significance. (The claim of increase in breaking strength can be supported.)]

3) A random sample of 50 items gives the mean 6.2 and standard deviation 10.24. Can it be regarded as drawn from a normal population with mean 5.4 at 5% level of significance?

   [Ans: $z_{cal} = 1.74$. Null hypothesis is accepted.]

4) The mean height of a random sample of 100 individuals from a population is 160. The Standard deviation of the sample is 10. Would it be reasonable to suppose that the mean height of the population is 165?

   [Ans: $z_{cal} = 5$, No]

5) The mean value of a random sample of 60 items was found to be 145, with a standard deviation of 40. Find the 95% confidence limits for the population mean. What size of the sample is required to estimate the population mean within 5 of its actual value with 95% or more confidence, using the sample mean?

   [Ans: $134.9 \le \mu \le 155.1$, least size of the sample is 246.]

2. Test of Significance of the difference between the means of two samples.

1) A simple sample of heights of 6400 English men has a mean of 170 cm and an SD of 6.4 cm, while a simple sample of heights of 1600 Americans has a mean of 172 cm and an SD of 6.3 cm. Do the data indicate that Americans are, on the average, taller than the Englishmen?

   [Ans: $z_{cal} = -11.32$, H_1 is accepted.(LOS 1%)]

2) Test the significance of the difference between the means of the samples, drawn from two normal populations with the same SD using the following data:

	Size	Mean	SD
Sample 1	100	61	4
Sample 2	200	63	6

   [Ans: $z_{cal} = -3.02$, H_1 is accepted. (LOS 5%)]

3) The average marks scored by 32 boys are 72 with SD of 8, while that for 36 girls is 70 with SD of 6. Test at 1% LOS whether the boys perform better than girls.

   [Ans: $z_{cal} = 1.15$, H_0 is accepted.]

4) An IQ test was given to a large group of boys in the age group of 18-20 years, who scored an average of 62.5 marks. The same test was given to a fresh group of 100 boys of the same age group. They scored an average of 64.5 marks with an SD of 12.5 marks. Can we conclude that the fresh group of boys has better IQ?

   [Ans: Yes. $z_{cal} = 1.6$]

5) The means of two simple samples of 1000 and 2000 items are 170 and 169 cm respectively. Can the samples be regarded as drawn from the same population with SD 10 at 5% LOS?

   [Ans: No. $z_{cal} = 2.58$]

6) Intelligence tests were given to two groups of boys and girls of the same age group chosen from the same college and the following results were obtained.

	Size	Mean	SD
Boys	100	73	10
Girls	60	75	8

Examine whether the difference between the means is significant or not.

[Ans: No. $z_{cal} = 1.32$]

7) In a college, 60 junior students are found to have a mean height of 171.5 cm and 50 senior students are found to have a mean height of 173.8 cm. Can we conclude, based on this data that the juniors are shorter than seniors at (i) 5% LOS and (ii) 1% LOS, assuming that the SD of the students of that college is 6.2 cm?

[Yes at 5% LOS, No at 1% LOS. $z_{cal} = 1.937$]

<u>3. Test of Significance of the single proportion.</u>

1) Experience has shown that 20% of manufactured product is of top quality. In one day's production if 400 articles, only 50 are of top quality. Show that wither the production of the day chosen was not a representative sample or the hypothesis of 20% was wrong.

[Ans: $z_{cal} = 3.75$. H_0 is rejected at 5% level of significance]

2) A salesman in a departmental stores claims that at most 60 percent of the shoppers entering the leaves without making purchase. A random sample of 50 shopper showed that 35 of them left without making a purchase. Are these sample results consistent with the claim of the salesman? Use an LOS of 0.05

[Ans: $z_{cal} = 1.443$. H_0 accepted at 5% level of significance]

3) In a particular area 984 girls against every 1000 boys. Can we say gender equality exist?

[Ans: $z_{cal} = -0.445$. H_0 accepted at 5% level of significance]

4) Certain crosses of pea gave 5321 yellow and 1804 green seeds. The expectation is 25% green seeds based on a certain theory. Is this divergence significant or due to sampling fluctuation?

[significant is due to sampling fluctuation?]

A company provided 1000 items are 3% defectively. A person pics up sample of 50 and he found 2 items defected. Can you say defectively has increased and he rejected entire slot. [Ans: $z_{cal} = 0.4145$, H_0 accepted ,No difference in defectivity]

4. Test of Significance of difference of proportion.

1) In a large city A, 20% of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5 percent of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant?

   [$z_{cal} = 0.92$, H_0 is accepted, The difference between proportion is not significant]

2) Before an increase in excise duty on tea, 800 people out of a sample of 1000 were consumers of tea. After the increase in duty, 800 people were consumers of tea in a sample of 1200 persons. Find whether there is significant decrease in the consumption of tea after the increase in duty.

   [$z_{cal} = 6.82$, H_0 is rejected, The difference between proportion is significant]

3) A Government has started giving tax incentives for installing certain energy saving devices. However, not entire population is aware about this offer. Out of 400 persons, selected randomly, 140 were 'unaware' about the offer. Construct interval estimation for 'proportion aware' about the offer with 95% confidence level.
   [0.6032,0.6967]

4) During a countrywide investigation, the incidence of TB was found to be 1%. In a college with 400 students, 5 are reported to be affected whereas in another college of 1200 students, 10 are found to be affected. Does this indicate any significance difference?

   [$z_{cal} = 0.725$, Not significant]

Small sample test

1. Tests made on the breaking strength of 10 pieces of a metal gave the following results: 578, 572, 570, 568, 572, 570, 570, 572, 596 and 584 kg. Test if the mean breaking strength of the wire can be assumed as 577 kg.

   [Ans. t=-0.65, H_0 accepted]

2. A machinist is expected to make engine parts with axle diameter of 1.75 cm. A random sample of 10 parts shows a mean diameter 1.85 cm with an SD of 0.1 cm. On the basis of this sample, would you say that the work of the machinist is inferior?

   [Ans. t=3, H_0 rejected]

1. Two independent samples of sizes 8 and 7 contained the following values:

Sample 1	19	17	15	21	16	18	16	14
Sample 2	15	14	15	19	15	18	16	

Is the difference between the sample means significant?

[Ans. t=0.93, H_0 accepted]

2. The following data represent the biological values of protein from cow's milk and buffalo's milk at a certain level

Cow's milk	1.82	2.02	1.88	1.61	1.81	1.54
Buffalo's milk	2.00	1.83	1.86	2.03	2.19	1.88

Examine if the average values of protein in the two samples significantly differ.

[Ans. t=-2.03, H_0 accepted]

3. The following data relate to the marks obtained by 11 students in 2 tests, one held at the beginning of a year and the other at the end of the year after intensive coaching.

Test 1	19	23	16	24	17	18	20	18	21	19	20
Test 2	17	24	20	24	20	22	20	20	18	22	19

Do the data indicate that the students have benefited by coaching?

[Ans. t=-1.38, H_0 accepted]

F test

1. A sample of size 13 gave an estimated population variance of 3.0 while another sample of size 15 gave an estimate of 2.5. Could both samples be from populations with the same variance?

[Ans. F=1.2, H_0 accepted]

2. Two random samples gave the following data:

Sample	Size	Mean	Variance
1	8	9.6	1.2
2	11	16.5	2.5

Can we conclude that the two samples have been drawn from the same normal population?

[Ans. F=2.007, H_0 accepted]

3. The nicotine contents in two random samples of tobacco are given below

Sample 1	21	24	25	26	27	
Sample 2	22	27	28	30	31	36

Can you say that the two samples came from the same population?

[Ans. F=4.07, H_0 accepted; t=-1.92, H_0 accepted]

Chi Square test

1. An unbiased doe is thrown 600 time out comes are given below. Can we really claim the die is unbiased?

1	2	3	4	5	6
110	98	92	103	105	92

[Ans: $\chi^2_{cal} = 11.07$, H_0 accepted at 5% LOS]

2. If one of the parent's blood group is A and B then blood group of child is A, AB, B past data shows A: AB: B=1:2:1. A sample of 250 new born baby with either parent's A is 70, B is 55 and remaining AB. Can we say at 5% LOS population is behaving expectedly?

A	70
AB	125
B	55

[Ans: $\chi^2_{cal} = 1.8$, H_0 accepted at 5% LOS]

3. The following data show defective article produced by 4 machines:

Machine	A	B	C	D
Production time	1	1	2	3
No. of defective	12	30	63	98

Do the figures indicate a significant difference in the performance of the machines?

[Ans: $\chi^2_{cal} = 11.82$, H_0 rejected at 5% LOS]

1. Two sample polls of voters for votes for two candidates A and B for a public office are taken, one from among the rural areas and another from urban areas. The results are in the following table

Area	Votes for	
	A	B
Rural	620	380
Urban	550	450

Examine whether the nature of the area is related to voting preference in this election.

[Ans: $\chi^2_{cal} = 10.092$, H_0 is rejected at 5% LOS]

2. A pharmaceutical company is considering vaccine for new disease

	Vaccinated	Not vaccinated
No. of affected people with disease	50	150
No. of not affected people with disease	25	125

Can we conclude that vaccination and disease are independent?

[Ans: $\chi^2_{cal} = 93.63$, H_0 is rejected at 5% LOS]

3. The following data is collected on two characters. Based on this, Can you say that there is no relation between smoking and literacy?

	Smokers	Non- smokers
Literates	83	57
Illiterates	45	68

[Ans: $\chi^2_{cal} = 9.22$, H_0 is rejected at 5% LOS]